When Does Deep Reinforcement Learning Help Link Adaptation? A Pre-Registered Evaluation Against Calibrated Rules

Author names (to be completed)

***Abstract*—**

[Draft v0 placeholder] Abstract (about 200 words). The scope sentence comes first. The warm-up, G6 and G10 clauses are selected by `\wordingsel` from the run results: for now, C2 reads "beats the OLLA family and the single-cue calibrated rules (calibrated on 10 to 20 traces)".

***Index Terms*—link adaptation, reinforcement learning, HARQ, evaluation methodology, pre-registration**

I. Introduction

[Draft v0 placeholder] Introduction (about 750 words), built from `outline.md` §I: motivation; the comparator gap (of 23 learning-based link-adaptation papers screened from abstracts and available text, 0 compared against a rule calibrated on the learner's own cues, 3 did so partly, and for 7 the text did not allow a determination); the study in one paragraph; the thesis; the four contributions (C1 protocol and artefacts; C2 a positive case, "beats the OLLA family and the single-cue calibrated rules (calibrated on 10 to 20 traces)"; C3 where RL does not help; C4 pitfalls and recommendations). Figure 2 is the thesis figure and is shown in Section VI in this draft.

II. Related Work

A. Classical link adaptation and OLLA

Outer-loop link adaptation (OLLA) corrects the reported channel quality indicator (CQI) with an offset that steps down after a NACK and up after an ACK, so that the long-run block-error rate approaches a target [1], [2]. Variants adapt the target [3], the step sizes [4], [5], [6], the offset estimator [7], [8], or the feedback used under URLLC soft-ACK [9]. Ahsen et al. use a rank-dependent multi-offset rule [10]; this is a different design from the Multi-OLLA comparator used here. SALAD varies three OLLA step sizes without learning [11].

B. Learned link adaptation

Bandit formulations choose the modulation and coding scheme (MCS) or the offset directly [12], [13], [14], [15]. Deep reinforcement learning (RL) has been applied with DQN, actor-critic, TD3 and PPO agents [16], [17], [18], [19], [20]. Deadline-aware RL on an OLLA loop is studied by Kela et al. [21], and learning on top of OLLA by Pulliyakode and Kalyanasundaram [22], Pang et al. [17] and Wang et al. (LOLLA) [23]. These works are evaluated in different settings from ours (LOLLA, for example, treats uplink with an immediate CRC outcome and no HARQ feedback delay), and we do not re-implement them. Our question is a different one: given a simulator with multi-process HARQ, how much of a learner's gain over OLLA remains once the comparison includes rules calibrated on the learner's own cues?

TABLE I
Eight closely related learned link-adaptation papers. "Calibrated rule": a non-learning rule calibrated on the learner's cues is a baseline. "Parity": the learner and that rule use comparable information. Entries come from the abstract and accessible text.

Paper	Setting	HARQ model	Calibr. rule	Parity
Saxena [13]	single link, AWGN and fading	first transmission only	no	partial
Ye [16]	LTE/NR simulation	not stated	no	unclear
Kela [21]	5G UL, XR, system-level	system-level HARQ	no	no
Pang [17]	LTE/NR simulation	not stated	no	unclear
You [18]	simulation and USRP/srsRAN testbed	parallel HARQ, ACK delay	partial	no
Wang (LOLLA) [23]	GPU NR stack, UL	immediate CRC, no HARQ delay	no	no
Demirel [20]	system-level, multi-cell	system-level	partial	unclear
Wiesmayr (SALAD) [11]	OAI, single user	real HARQ	partial	yes

Table I summarises eight closely related papers on two points that matter for that question. Across the 23 learning-based papers we screened, 0 compared against a calibrated rule built on the learner's cues, 3 did so partly, 13 did not, and 7 could not be determined. These counts rest on abstracts and the text we could access, and an "unclear" entry is not a negative finding.

C. Evaluation rigour

Doherty et al. show that carefully tuned heuristics matched published RL results on optical-network problems [24]; this is the closest methodological precedent, in another domain. General concerns about reproducibility and baselines in learned systems and in wireless machine learning are raised in [25], [26], [27], [28]. For RL specifically, variance across seeds and the reporting of intervals are discussed in [29], [30], [31]. Pre-registration is advocated in [32], [33], and credibility of simulation studies in [34], [35]. We apply these ideas to one concrete question in link adaptation.

Draft v0. Sections marked as placeholders wait on runs or on later drafting.

III. System Model

All results come from one MATLAB simulator, evaluated in simulation only. Table II lists the parameters, and each is checked against the code by the number-generation script.

A. Cell and channel

A single macro cell serves $N = 10$ users on a $20\,\mathrm{MHz}$ carrier at $2\,\mathrm{GHz}$ with 100 resource blocks (RBs), $15\,\mathrm{kHz}$ subcarrier spacing and a $1\,\mathrm{ms}$ transmission time interval (TTI). Large-scale fading is log-normal shadowing with $\sigma = 6$ dB and 50 m decorrelation distance; small-scale fading is Jakes. User speeds are drawn from 3–30 m/s, which gives Doppler frequencies of about $20\,\mathrm{Hz}$ to $200\,\mathrm{Hz}$. A two-state Markov process adds interference bursts of 9 dB with mean dwell times of 250 TTIs (calm) and 120 TTIs (burst).

B. CQI feedback and link abstraction

Each user reports a 15-level channel quality indicator (CQI) with a delay drawn from 2–8 TTIs and Gaussian report noise of $\sigma = 1$ dB. Block errors follow an SINR-to-BLER abstraction rather than LDPC link-level tables; this is a limitation (Section VIII). The OLLA reference loop adds an offset Δ to the reported CQI. A NACK lowers Δ by $\Delta_{\downarrow} = 0.5$ dB, and an ACK raises it by $\Delta_{\uparrow} = \Delta_{\downarrow} P_{\mathrm{tgt}}/(1 - P_{\mathrm{tgt}}) \approx 0.056$ dB, for a target $P_{\mathrm{tgt}} = 0.1$, with $|\Delta| \leq 10$ dB. The corrected CQI selects the MCS. We call this corrected CQI the *OLLA reference*. All calibrated rules and the learned action act as CQI-index offsets on top of it.

C. HARQ

Retransmissions use chase combining ($2.5\,\mathrm{dB}$ gain per retransmission, at most 3 retransmissions) and reuse the stored MCS and transport block size. The learner's action therefore affects new transmissions only. We study two timing regimes: *realistic* HARQ with an ACK/NACK round-trip time (RTT) of 8 TTIs and 8 processes per user, and *instantaneous* feedback (RTT 0, one process).

D. Scheduler

A proportional-fair (PF) scheduler gives all RBs to one user per TTI. The PF average rate uses an exponential average with $\alpha = 0.05$ and is updated *at scheduling time* with the scheduled rate. Updating it instead when the ACK arrives is a common alternative; under multi-process HARQ it changes conclusions (Section VII), so we state the choice explicitly (Fig. 1b).

E. Traffic and metrics

Deadline traffic: $12\,000\,\mathrm{bit}$ packets arrive every 20 TTIs with a budget of 10 TTIs. The primary metric is the on-time delivery ratio. *Full-buffer traffic:* the primary metric is goodput. Each trace lasts 200 TTIs, of which the first 50 are burn-in, leaving 150 scored TTIs. We also report the Jain fairness index over per-user means and the 5th-percentile user value (edge5). Four cases form the main campaign: M1 (deadline, RTT 8), M2 (deadline, RTT 0), M3 (goodput, RTT 8) and M4 (goodput, RTT 0).

TABLE II
Simulation parameters (checked against `defaultEnv_v17.m` and `v17_setup.m`).

Parameter	Value
Users, bandwidth, RBs	10, 20 MHz, 100
Carrier, subcarrier spacing, TTI	2 GHz, 15 kHz, 1 ms
Shadowing	6 dB, 50 m
Speed range (Doppler)	3–30 m/s (20–200 Hz)
Interference burst	9 dB; dwell 250/120 TTI
CQI delay, noise	2–8 TTI, 1 dB
OLLA target, steps, limit	0.1, 0.5/0.056 dB, 10 dB
HARQ	chase, 2.5 dB, ≤ 3 retx; RTT 8/8 proc. or RTT 0/1 proc
PF averaging	scheduling time, $\alpha = 0.05$
Deadline traffic	12000 bit / 20 TTI, budget 10 TTI
Trace length, burn-in	200, 50 TTI
Evaluation traces	500 per case

IV. Learning Formulation and Method

A. Decision problem

At every TTI each scheduled user receives one decision. The effective CQI is $\mathrm{clip}(q_{\mathrm{ref}} + b_i + \delta)$, where q_{ref} is the OLLA reference, b_i is a calibrated base-rule offset and δ is the learned part. In the *offset* formulation $b_i = 0$ and $\delta \in [-4, +2]$. In the *residual* formulation $\delta \in \{-1, 0, +1\}$ and the base rule is the better of Fixed and FixedPU, chosen on calibration traces (it was FixedPU in all four cases). FixedPU uses the true delay and speed class of each user and is therefore a *class oracle* (Section V).

B. Rewards

For deadline traffic the reward is $+1$ for an on-time and -1 for a missed packet, with potential-based shaping $\Phi(s) = -0.5\, r/B$ for r remaining bits of a B-bit packet [36] and a deviation penalty of $0.02|\delta|$. For goodput the reward is an opportunity cost, the fix for the slot-stealing failure in Section VII.

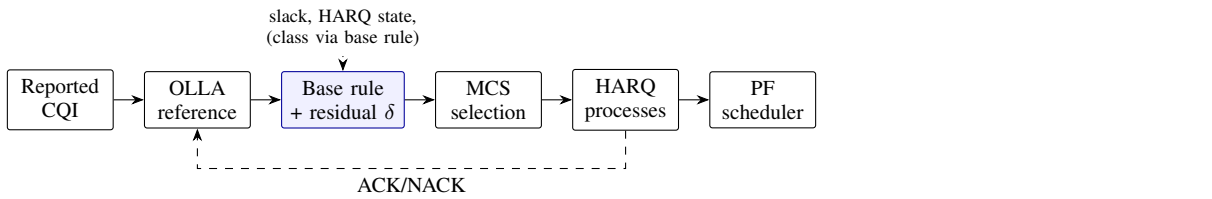
C. Learner and training

A shared multi-user advantage actor-critic (A2C) [37] uses a 57-128-96-3 actor on three stacked frames of 19 features. It has 20 099 parameters and 19 872 multiply-accumulates per user decision. The logit of $\delta = 0$ starts with a bias of 3. We train for 3000 episodes (deadline) or 800 (goodput) with 10 independent seeds. The *primary scheme* is the per-seed mean: each seed's policy is evaluated and the results are averaged. A deterministic ensemble of the seeds is secondary (pre-registered from v14). Hyper-parameters are in Appendix A. Wall-clock inference time was not measured; the count above is analytic.

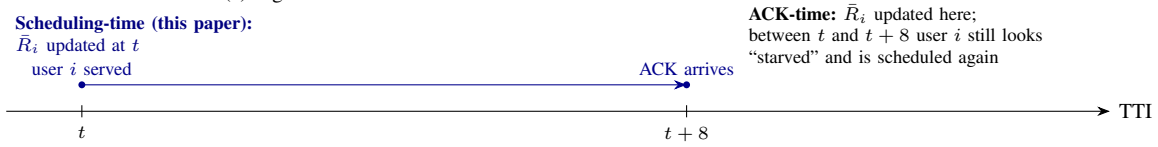
V. Evaluation Protocol

A. Cases and registry

Each round of the campaign uses fresh, disjoint seeds and traces: 500 evaluation traces and 10 training seeds for the headline cases. Every case is recorded in a public registry of 67 cases and 108 scheme evaluations (v9 to v17, Aug–Nov 2026). Of these, 7 cases (18 evaluations) were later superseded



(a) Signal flow



(b) PF average timing at RTT 8

Fig. 1. (a) Decision path: the learner (shaded) acts as a bounded offset on a calibrated base rule applied to the OLLA reference CQI; the dashed arrow is the OLLA loop. (b) The two PF-average definitions on an RTT-8 timeline.

after an error was found (Section VII), and 49 cases date from v10, when the present protocol began.

B. Comparators, calibration and information

Nine comparators are evaluated (seven distinct ones in M1, because the tuned-target OLLA equals the default OLLA and the two bandits coincide there). *Baseline*: open-loop CQI. *OLLA*, *OLLA (tuned target)* and *Multi-OLLA* (a fast/slow blend). *Fixed*: one global offset on the OLLA reference. *FixedPU*: one offset per user class on the OLLA reference, where the class is the true delay class (two) and the true mean-speed class (three); this is a *class oracle*, because the true speed is not observable at the gNB. *Context table*: one offset per remaining-slack bin (three bins). Two contextual bandits (global; per delay class) are trained on the learner's episodes.

Table III states what each scheme observes. The learner's state contains only gNB-observable quantities and adds the slack cue and three HARQ-process features. But the residual policy acts on top of FixedPU, so it inherits the class information of that base rule, including the oracle speed class. For this reason we describe the calibrated rules as *built on the learner's cues* (the slack cue in the Context table, the class information in FixedPU). A rule that combines slack, class and HARQ-process state is the subject of the G6 comparator (Section VI-G).

C. Calibration of the comparators: three findings

A code review before the final comparison showed that the v17 calibration of the rules was weaker than a "calibrated on the same cues" description suggests. We state three findings plainly.

1) *The v17 Context table was found by a two-pass coordinate ascent started from the all-zero table, on 10 calibration traces.* The search space has $7^3 = 343$ tables (seven offsets in three bins), and the search did not examine all of them.
2) *The v17 FixedPU table was separable.* For each of the nine global offsets one run was made, and each of the six classes took its own best offset from the per-class metrics. The assembled six-entry table was never evaluated as a whole table.
3) *A pre-registered correction (G6) removes both limits.* Fixed (9 candidates), the Context table (343 tables) and the tuned-target OLLA (30 settings) are searched exhaustively on 500 fresh calibration traces (seeds 600001–600500). FixedPU is a separable grid followed by joint refinement (at most three sweeps). A combined table over slack $\times$ delay class $\times$ speed class $\times$ busy HARQ processes (0 or ≥ 1; $3 \cdot 2 \cdot 3 \cdot 2 = 36$ cells) is searched by coordinate ascent from four starts. The first start is the *composite*

$$\mathrm{composite}(b,d,v,u) = \min\big(2, \max(-4,\ P(d,v)+X(b)-F)\big),$$

with F the re-calibrated Fixed offset, $P(d,v)$ the re-calibrated FixedPU table and $X(b)$ the re-calibrated Context table, all in integer CQI steps. Cells with fewer than 200 calibration decisions are not searched and keep their composite value. The tables are hashed before any evaluation trace is run, and calibration and evaluation values are reported side by side.

The G6 outcomes are pending at the time of writing; the M1 results in Section VI use the v17 tables of findings 1 and 2.

D. Splits, common random numbers and decision rules

Calibration traces (seeds from 60001), validation traces (from 80001) and evaluation traces (from 500001) are disjoint. All schemes see the same evaluation traces with common random numbers [38], so comparisons are paired per trace. "Scheme A beats comparator X" means a Holm-adjusted [39] two-sided Wilcoxon signed-rank $p < 0.05$ [40] *and* a lower bound > 0 for the 90 % confidence interval of the paired percentage margin. *Claim A*: beats the OLLA family. *Claim B*: beats Fixed, FixedPU and the Context table. A *WIN* means beating all nine comparators. A fairness guard applies: a Jain-index drop above 0.02 or an edge5 drop above 10 % relative to the best calibrated rule must be reported. Two headroom bounds frame the results: a per-trace hindsight bound and a genie bound.

TABLE III
INFORMATION USED BY EACH SCHEME. ✓: USED; –: NOT USED; "ORACLE": SIMULATOR GROUND TRUTH, NOT A GNB MEASUREMENT.

Scheme	CQI	ACK hist.	Slack	Delay cls.	Speed cls.	HARQ state	Other	Calibrated on
Baseline (open loop)	✓	–	–	–	–	–	–	–
OLLA	✓	✓	–	–	–	–	–	– (standard)
OLLA, tuned target	✓	✓	–	–	–	–	–	target grid, 10 traces
Multi-OLLA	✓	✓	–	–	–	–	CQI trend, cell NACK average	–
Fixed	✓	✓	–	–	–	–	–	offset grid, 20 traces
FixedPU (class oracle)	✓	✓	–	✓	✓ *oracle*	–	–	per-class grid, 20 traces
Context table	✓	✓	✓(3 bins)	–	–	–	–	coord. ascent, 10 traces
Bandit (global)	✓	✓	–	–	–	–	–	learner's episodes
Bandit (delay class)	✓	✓	–	✓	–	–	–	learner's episodes
A2C, offset action	✓	✓	✓	–	–	✓	ΔCQI, observed SINR, OLLA offset, 3 frames	800 ep. × 10 seeds
Residual A2C (primary)	✓	✓	✓	✓ (base)	✓ *oracle (base)*	✓	as offset A2C	3000 ep. × 10 seeds

Trace-level intervals treat seeds as fixed. We therefore also report seed-level inference: a hierarchical seed×trace bootstrap [41] (20,000 resamples), a bootstrap over the ten per-seed means, and a one-sided sign test over seeds. Under the pre-registered rule, a claim that passes at trace level but whose seed-level lower bound is not above zero is reported as "not robust to seed resampling".

E. Pre-registration and multiplicity

From v12 on, a pre-registration note was written before each run, and failed predictions are reported. In v17 the primary predictions were met in all four cases, and two secondary predictions on M1 were not: the 800-episode offset policy was predicted to win and failed Claim B, and the 800-episode residual policy was predicted to fail Claim B and won. Across the registry, 9 of the 60 valid cases show a WIN under the corrected verdict. Multiplicity is handled over the whole registry: the M1 comparison against the best rule has Holm-adjusted $p = 9.1 \times 10^{-13}$, and $p = 8.8 \times 10^{-10}$ after a Bonferroni factor of 972 (108×9 comparators). We state plainly that the recipe was chosen during the campaign; M1 is a confirmatory test of that recipe on fresh data.

VI. RESULTS

A. Deadline traffic with realistic HARQ (M1)

The residual A2C policy (primary scheme) delivers 61.26 % of packets on time in M1, against 60.31 % for the Context table, 60.14 % for Fixed, 59.65 % for FixedPU and 54.72 % for OLLA (Fig. 2, Table IV). Its margin is +11.96 % over OLLA, +12.16 % over Multi-OLLA, +2.70 % over FixedPU, +1.87 % over Fixed and +1.58 % over the Context table (90 % CI [+1.23, +1.94] %; Holm $p = 9.1 \times 10^{-13}$). The policy beats all nine comparators under the rule of Section V. In absolute terms the margin over the Context table is +0.95 percentage points. The tuned-target OLLA coincides with the default OLLA in M1 (target 0.10 was selected), and the two bandits coincide.

How much of the gain over OLLA do simple calibrated rules already capture? A single calibrated offset (Fixed) captures 83 % of the learner's gain over OLLA, and the Context table captures 85 %. The comparison against OLLA alone would therefore overstate what the learner adds.

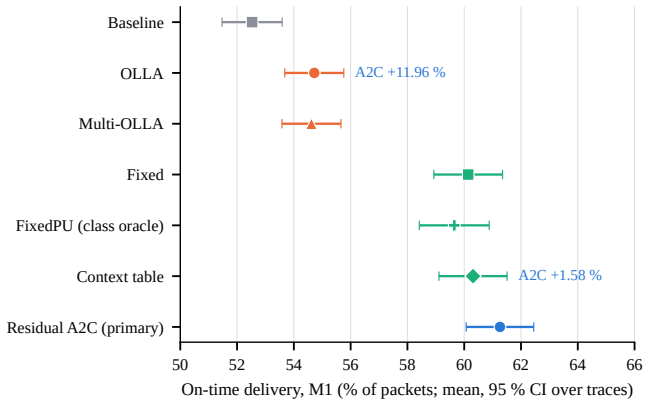


Fig. 2. M1 on-time delivery ratio by scheme (mean and 95 % CI over 500 traces). Blue: residual A2C; orange: OLLA family; green: calibrated rules; gray: baseline. Annotations give the margin of the primary scheme.

B. Seed-level robustness

Trace-level intervals treat the ten training seeds as fixed. Seed-level views are therefore reported next to them (Fig. 4). Against the Context table, 9 of 10 seeds lie above zero; the exception is seed 2 (−0.80 %), and the one-sided sign test gives $p = 0.0107$. The 95 % interval for the mean margin is [+0.72, +2.53] % by bootstrap over the ten seed means, [+0.48, +2.68] % by a t-interval, and [+0.62, +2.62] % by the hierarchical seed×trace bootstrap. Against Fixed the corresponding intervals are [+1.01, +2.81] %, [+0.77, +2.98] % and [+0.89, +2.90] %. All seed-level lower bounds are above zero, so the pre-registered "not robust to seed resampling" qualifier does not apply to M1. The hierarchical 90 % interval is about 2.3 times wider than the trace-only one, which is the price of counting seed variance.

C. Fairness and cost

The gain does not come at the expense of fairness under the guard. Against the Context table the Jain index differs by −0.0025 (0.7715 against 0.7740), and edge5 is +1.2 % higher (0.2148 against 0.2122). Goodput is 4.4506 Mbps against 4.3635 Mbps.

TABLE IV
M1 COMPARATOR TABLE. MARGIN: PAIRED PERCENTAGE GAIN OF RESIDUAL A2C OVER THE COMPARATOR, WITH THE 90 % CI OVER TRACES AND THE HOLM-ADJUSTED WILCOXON p. "BEATS": THE PRE-REGISTERED RULE IS MET.

Comparator	On-time (%)	Margin (%)	90 % CI (%)	Holm p	Beats
Baseline	52.53	+16.62	[+15.74, +17.50]	2.8×10^{-76}	yes
OLLA	54.72	+11.96	[+11.21, +12.71]	1.2×10^{-66}	yes
OLLA (tuned target)	54.72	+11.96	[+11.21, +12.71]	1.2×10^{-66}	yes
Multi-OLLA	54.62	+12.16	[+11.40, +12.92]	3.1×10^{-67}	yes
Bandit (global)	41.80	+46.55	[+45.22, +47.88]	1.3×10^{-82}	yes
Bandit (delay class)	41.80	+46.55	[+45.22, +47.88]	1.3×10^{-82}	yes
Fixed	60.14	+1.87	[+1.52, +2.23]	1.3×10^{-16}	yes
FixedPU (class oracle)	59.65	+2.70	[+2.36, +3.03]	1.3×10^{-31}	yes
Context table	60.31	+1.58	[+1.23, +1.94]	9.1×10^{-13}	yes

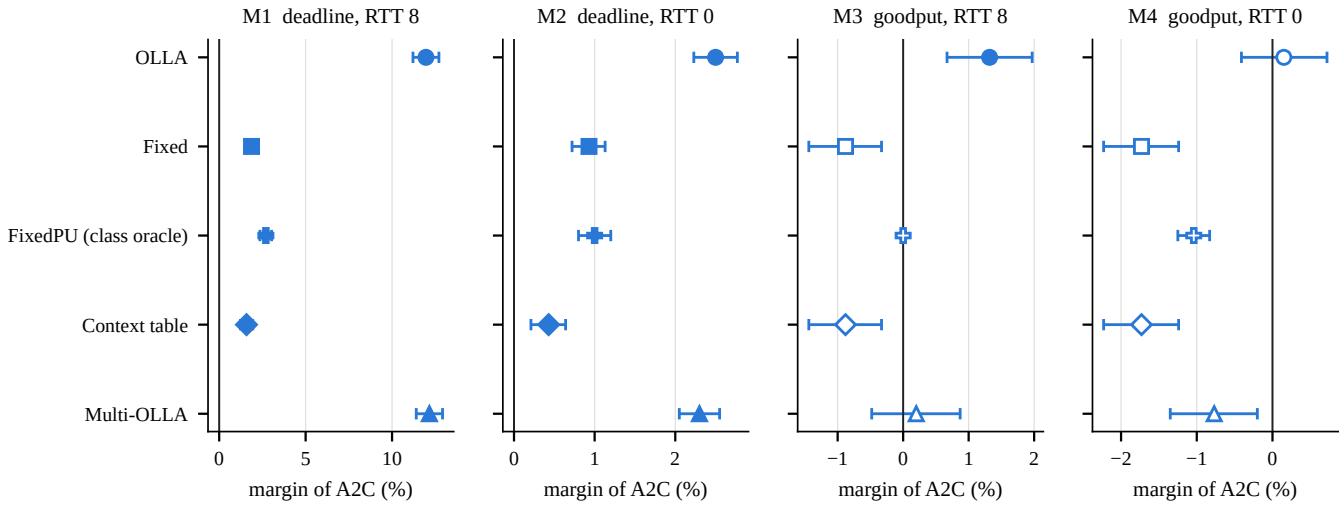


Fig. 3. Margin of residual A2C over each comparator in the four cases (90 % CI over traces). Filled markers: the pre-registered "beats" rule is met; open markers: it is not. The two bandits are omitted for scale (see Table IV).

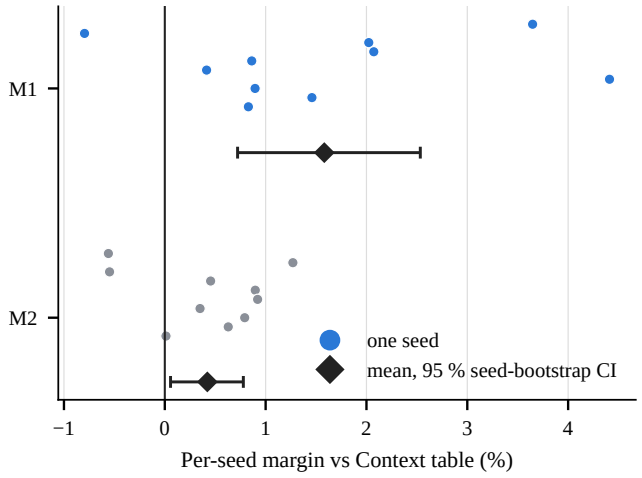


Fig. 4. Per-seed margin over the Context table for M1 and M2 (one dot per training seed) with the mean and its 95 % bootstrap interval over seed means (diamond).

D. Replication without HARQ delay (M2)

With instantaneous feedback (M2) the margin over the Context table is +0.43 % (90 % trace interval lower bound +0.21 %), and the hierarchical 95 % interval is [−0.03, +0.87] %, so M2 is described as a replication only: 8 of ten seeds lie above zero. Earlier RTT-0 wins (v13, v14, v16) used the other PF-averaging definition and are not used as evidence.

E. Goodput: no gain (M3, M4)

On full-buffer goodput the primary scheme does not beat calibrated rules: −0.88 % (M3) and −1.73 % (M4) against Fixed. At RTT 8 the residual policy is bit-identical to its base rule in 5000 of 5000 seed×trace cells, which means that the policy learned to set $\delta = 0$. At RTT 0 this is not the case (2985 of 5000 cells are identical), and the policy does worse than its base rule.

F. Headroom and training length

How much room is there? The per-trace hindsight bound over Fixed is +3.4 % in M1 (+2.0 % in M2), and a genie bound +37.8 % (Fig. 5a). The learner's M1 margin of +0.95 points is thus bounded by the information in the state, not by training

alone, but training length matters: at 800 episodes the margin over Fixed is +0.62 % (90 % CI [+0.34, +0.91] %) and at 3000 episodes it is +1.87 % ([+1.52, +2.23] %) (Fig. 5b). The 800-episode variant passes Claim B at trace level but only 6 of ten seeds lie above the Context table.

G. Robustness checks

[Draft v0 placeholder] Four pre-registered checks wait on runs: W1 (OLLA warm-up), G6 (re-calibrated and combined comparators; verdict: beats the OLLA family and the single-cue calibrated rules (calibrated on 10 to 20 traces)), NR1 (non-oracle base rule; The M1 result uses a class-oracle base rule; a non-oracle base was not tested at RTT 8.) and G10 (two neighbouring operating points; The result is established at one operating point (12 kbit packets, 10-TTI budget); other loads and deadlines were not tested under the corrected scheduler.). Run status: W1 pending, G6 pending.

H. Policy behaviour

[Draft v0 placeholder] The action re-cut (G1) has not been run. No mechanism is claimed for the gain.

VII. Pitfalls and Recommendations

Three simulation or design errors changed conclusions during the campaign. We describe them because each can recur in other link-adaptation studies.

A. Mobility time scale

Early versions (up to v9) moved users v metres per TTI instead of $v\,T_{\mathrm{tti}}$. Channel dynamics were then about a thousand times too fast. Under this error the learner appeared to beat OLLA; after the correction, the v9 registry shows 0 wins in 18 cases, and the A2C policy was -4.08 % against OLLA on goodput.

B. Slot stealing

A per-user goodput reward let a shared agent lower one user's rate to benefit others' slots. The learned policy was bimodal (58 % of actions at -3 and 35.8 % at $+2$), and goodput was 1.38 Mbps against 2.92 for OLLA. An opportunity-cost reward removed the effect (2.9191 against 2.9233 Mbps, a -0.14 % difference that is not significant).

C. PF-average timing under multi-process HARQ

If the PF average is updated with ACKed bits at $t+\mathrm{RTT}$, a just-served user looks starved for RTT TTIs and is scheduled again. In a six-trace diagnostic, Baseline goodput at RTT 8 rose from 3.466 to 4.837 Mbps when going from one to eight processes (+39.6 %) while the Jain index fell from 0.503 to 0.420. Re-running all 13 schemes with the other PF definition changed goodput by 10–26 %, but that comparison is confounded by traces, drain and useful-bit accounting, so it is quoted only as a range. A 10.71 % engine discontinuity had the same cause. The sign of a simple comparison depends on the definition: open-loop Baseline was +4.5 % above OLLA under delivered-throughput PF (v9) but OLLA is +7.4 % above Baseline under scheduling-time PF in M3 (and +6.9 % in M4). A controlled one-trace-set demonstration (A6) is pending.

D. Withdrawn and superseded results

Results from v7 and v8 are withdrawn. A v10 exploratory win (E4c) used overlapping traces. The v12 pre-registered replication failed Claim B and the v13 X1 prediction failed. The v15 and v16 RTT>0 cases (7 cases, 18 evaluations) are superseded because they used the ACK-time PF average. The registry in the supplement lists every case.

E. Recommendations

1) Compare against rules calibrated on the learner's cues, state the calibration budget, and give an information table such as Table III.
2) Use common random numbers, disjoint calibration, validation and evaluation traces, and a public case registry.
3) Report seed-level intervals next to trace-level ones.
4) Report results by mobility regime, HARQ timing and observation set. The observation-set ablation in LOLLA (Table V(a) of [23], 254, 185 and 171 Mbps) shows how much a result can depend on what the learner sees.
5) State the PF-averaging definition and report the complexity of the learner.

VIII. Discussion and Limitations

[Draft v0 placeholder] Discussion (about 650 words): when RL helps (a decision with structure that a static offset cannot use, anchored to a calibrated base rule; no gain on full-buffer goodput), the practical reading (calibrate first; the residual adds about 1–2 % on-time at the studied point), and the limitations: one cell with $N = 10$ users and one user per TTI; SINR-to-BLER abstraction without LDPC tables or an ns-3 validation; 150 scored TTIs and a cold-start OLLA; one operating point (The result is established at one operating point (12 kbit packets, 10-TTI budget); other loads and deadlines were not tested under the corrected scheduler.); FixedPU is a class oracle that the residual policy inherits (The M1 result uses a class-oracle base rule; a non-oracle base was not tested at RTT 8.); calibration-budget asymmetry (G6); seed variance and selection over 17 versions; PF-definition dependence; no learned baseline from the literature re-implemented; one MATLAB code base.

IX. Conclusion

[Draft v0 placeholder] Conclusion (about 100 words), then code and data availability, pre-registration availability, the AI-assistance disclosure in the acknowledgments and the author statements.

References

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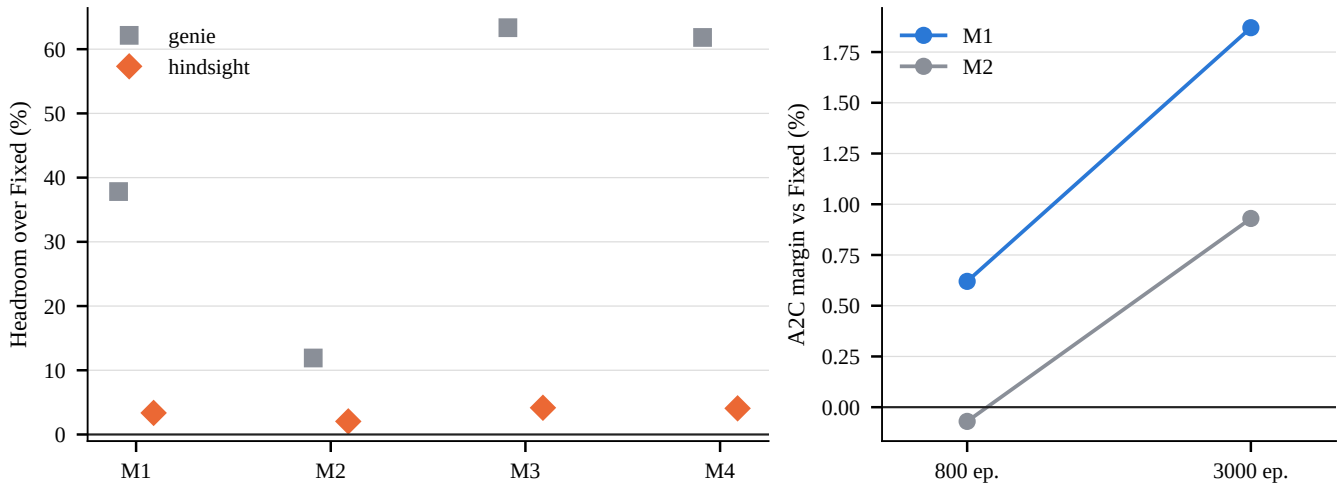


Fig. 5. (a) Headroom over Fixed in the four cases: genie and per-trace hindsight bounds. (b) Margin of residual A2C over Fixed after 800 and 3000 training episodes.

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